

LAB EXPERIMENTS

TOOL:R Programming

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COURSE:Data Handling and Visualization

Code: DSA0604

SET-1: Monthly Sales Data

R PROGRAM:

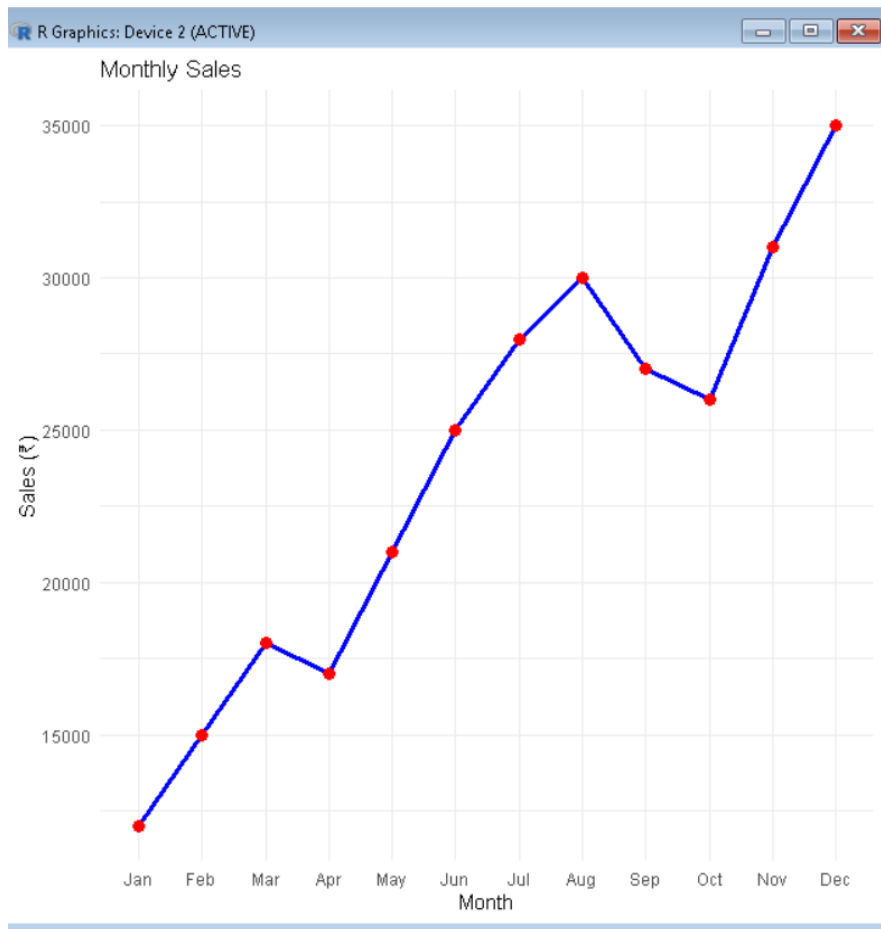
Question 1: Line Chart for Monthly Sales

```
library(ggplot2)

# Sample Dataset
sales <- data.frame(

  Month = factor(month.abb, levels = month.abb),
  Sales = c(12000,15000,18000,17000,21000,25000,
            28000,30000,27000,26000,31000,35000)
)

# Line Chart
ggplot(sales, aes(x = Month, y = Sales, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Monthly Sales",
    x = "Month",
    y = "Sales (₹)"
  ) +
  theme_minimal()
```



Q2: Bar Chart for Top-Selling Products

R program:

```
library(ggplot2)
```

```
products <- data.frame(
```

```
  Product = c("Laptop","Mobile","Tablet","Headphones","Smart Watch"),
```

```
  Sales = c(500,800,350,600,450)
```

```
)
```

```
ggplot(products,
```

```
  aes(x = reorder(Product, Sales),
```

```
    y = Sales,
```

```
    fill = Product)) +
```

```
geom_bar(stat = "identity") +
```

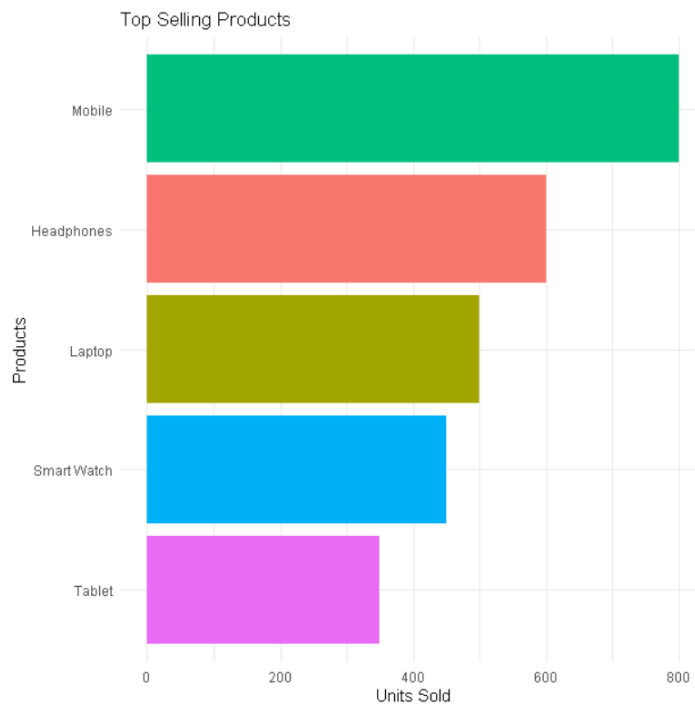
```
coord_flip() +
```

```
labs(
```

```

title = "Top Selling Products",
x = "Products",
y = "Units Sold"
) +
theme_minimal() +
theme(legend.position = "none")

```



Q3: Scatter Plot (Advertising Budget vs Monthly Sales)

R program:

```
# Question 3: Scatter Plot
```

```
library(ggplot2)
```

```
marketing <- data.frame(
```

```
  Advertising = c(5,7,8,10,12,15,18,20,22,24,27,30),
```

```
  Sales = c(12000,15000,17000,18000,21000,25000,
            27000,29000,30000,32000,34000,36000)
```

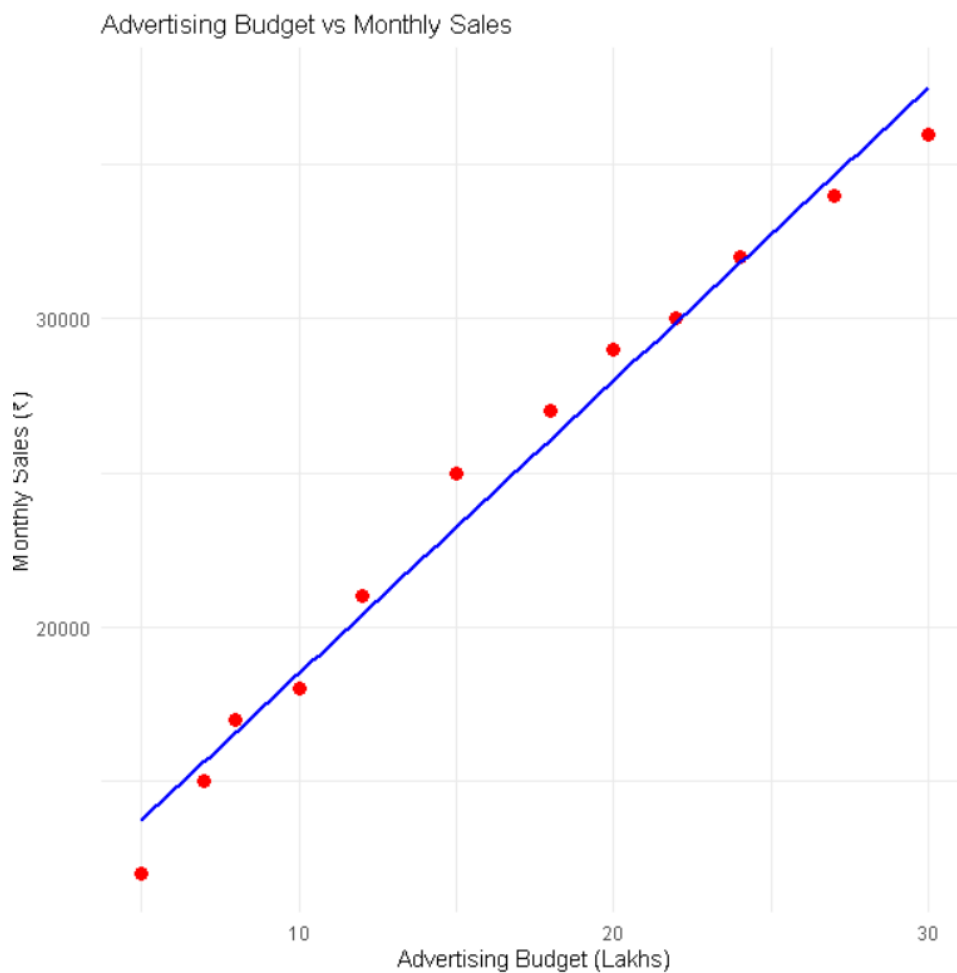
```
)
```

```
ggplot(marketing,
```

```
  aes(x = Advertising,
```

```
      y = Sales)) +
```

```
geom_point(color = "red", size = 3) +
geom_smooth(method = "lm",
            color = "blue",
            se = FALSE) +
labs(
  title = "Advertising Budget vs Monthly Sales",
  x = "Advertising Budget (Lakhs)",
  y = "Monthly Sales (₹)"
) +
theme_minimal()
```



Q4: Interactive Dashboard (Line Chart + Bar Chart)

R program:

CODE :

Install packages if needed

```

# install.packages("flexdashboard")

# install.packages("plotly")


library(flexdashboard)
library(plotly)
library(ggplot2)


# Dataset 1
sales <- data.frame(
  Month=factor(c("Jan","Feb","Mar","Apr","May","Jun",
    "Jul","Aug","Sep","Oct","Nov","Dec"),
    levels=c("Jan","Feb","Mar","Apr","May","Jun",
    "Jul","Aug","Sep","Oct","Nov","Dec")),
  Sales=c(12000,15000,18000,17000,22000,25000,
    24000,27000,30000,32000,35000,40000)
)

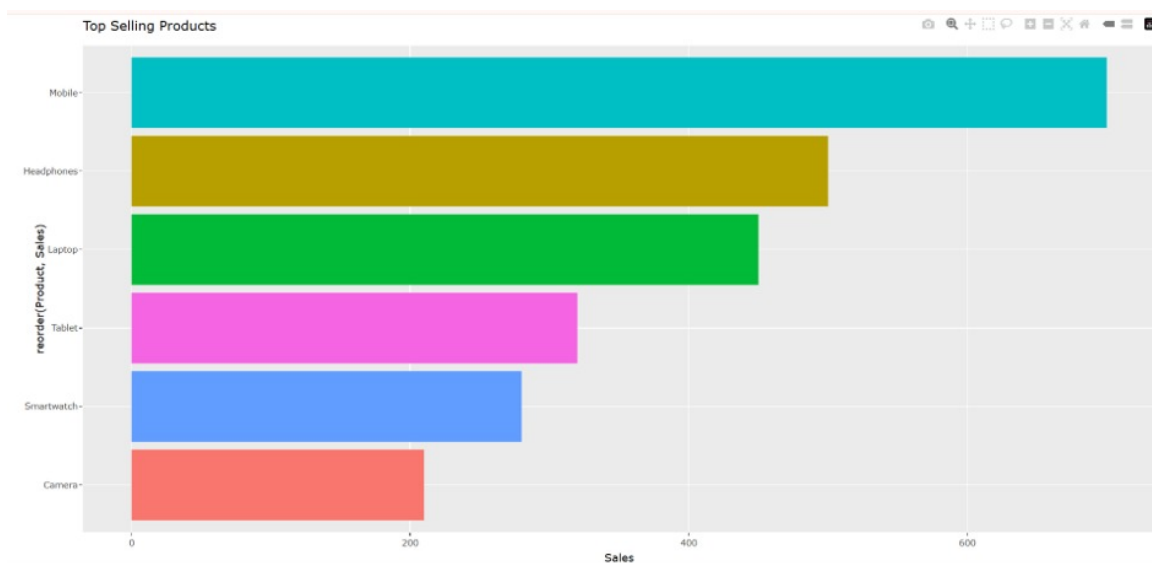
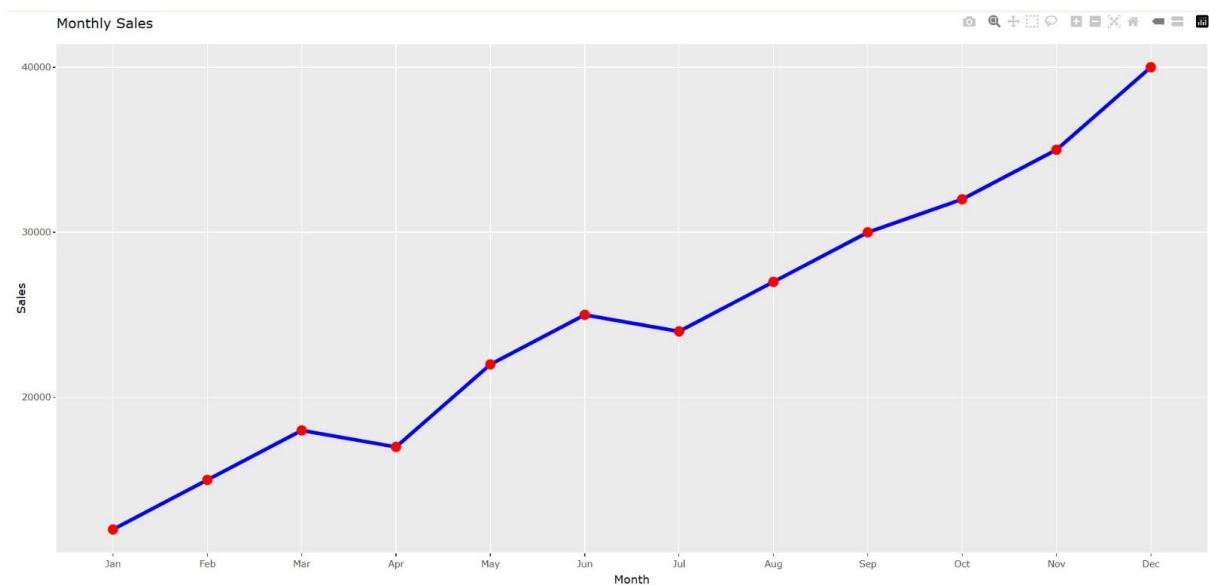

# Dataset 2
products <- data.frame(
  Product=c("Laptop","Mobile","Tablet","Headphones",
    "Smartwatch","Camera"),
  Sales=c(450,700,320,500,280,210)
)


# Line Chart
line_plot <- ggplot(sales, aes(Month, Sales, group=1)) +
  geom_line(color="blue", linewidth=1.2) +
  geom_point(size=3, color="red") +
  labs(title="Monthly Sales")

```

Bar Chart

```
bar_plot <- ggplot(products,  
  aes(reorder(Product, Sales), Sales, fill=Product)) +  
  geom_bar(stat="identity") +  
  coord_flip() +  
  labs(title="Top Selling Products") +  
  theme(legend.position="none")  
ggplotly(line_plot)  
ggplotly(bar_plot)
```



SET2: Customer Feedback Analysis

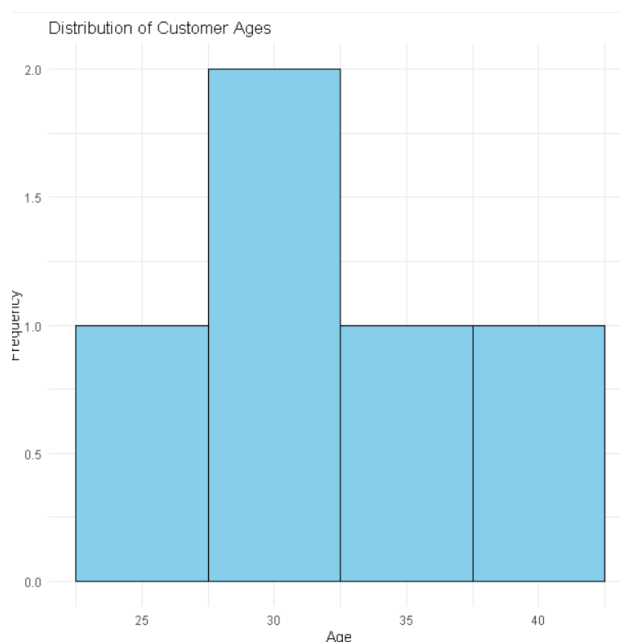
Q1:Histogram of Customer Ages

R PROGRAM:

```
library(ggplot2)

customer <- data.frame(
  CustomerID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction = c(4,5,3,4,5)
)

ggplot(customer, aes(x = Age)) +
  geom_histogram(binwidth = 5,
    fill = "skyblue",
    color = "black") +
  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Frequency"
  ) +
  theme_minimal()
```



Q2: Pie Chart of Customer Satisfaction Scores

Question 2: Pie Chart

R PROGRAM:

```
library(ggplot2)

# Dataset

customer <- data.frame(
  CustomerID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction = c(4,5,3,4,5)
)

# Count Satisfaction Scores

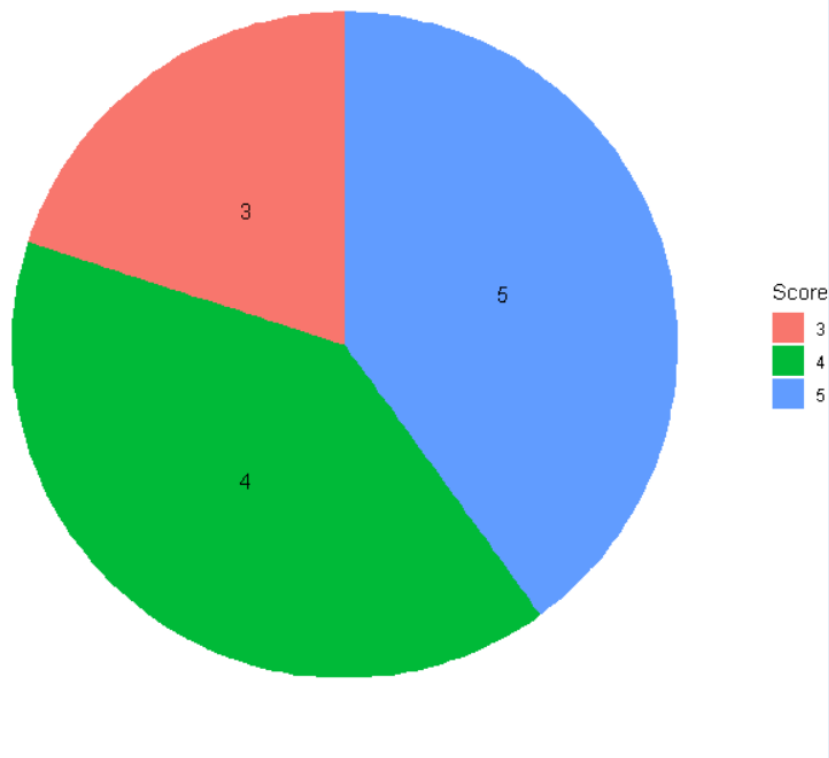
satisfaction_count <- as.data.frame(table(customer$Satisfaction))

colnames(satisfaction_count) <- c("Score", "Count")

# Pie Chart

ggplot(satisfaction_count,
  aes(x = "", y = Count, fill = Score)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  geom_text(aes(label = Score),
    position = position_stack(vjust = 0.5)) +
  labs(
    title = "Customer Satisfaction Scores"
  ) +
  theme_void()
```


Customer Satisfaction Scores



Q3- Stacked Bar Chart by Age Group

R PROGRAM:

Question 3: Stacked Bar Chart

```
library(ggplot2)
```

```
# Dataset
```

```
customer <- data.frame(  
  CustomerID = c(1,2,3,4,5),  
  Age = c(25,30,35,28,40),  
  Satisfaction = c(4,5,3,4,5)  
)
```

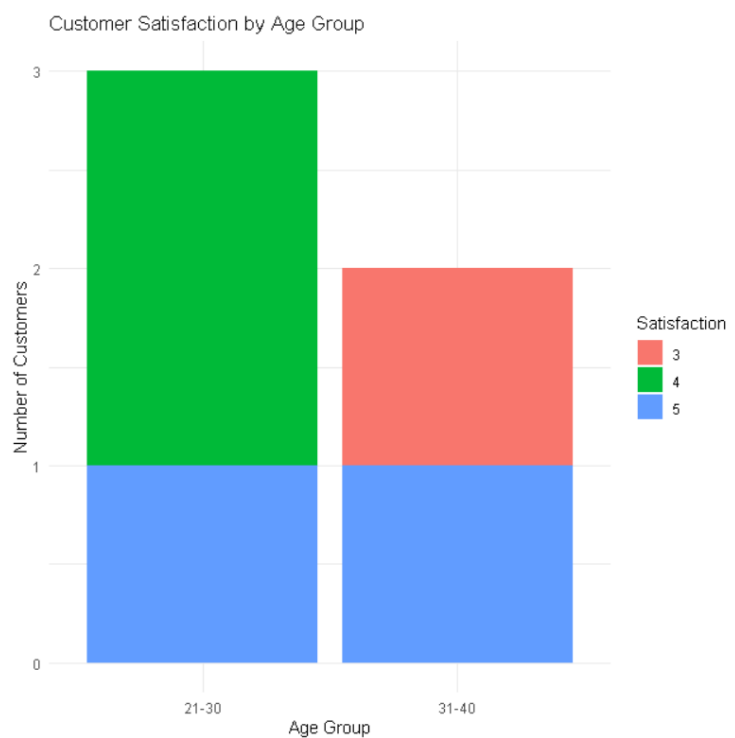
```
# Create Age Groups
```

```
customer$AgeGroup <- cut(  
  customer$Age,  
  breaks = c(20,30,40,50),  
  labels = c("21-30", "31-40", "41-50"),
```

```

include.lowest = TRUE
)
# Stacked Bar Chart
ggplot(customer,
  aes(x = AgeGroup,
      fill = factor(Satisfaction))) +
geom_bar() +
labs(
  title = "Customer Satisfaction by Age Group",
  x = "Age Group",
  y = "Number of Customers",
  fill = "Satisfaction"
) +
theme_minimal()

```



Q4: Word Cloud from Customer Feedback

R program:

```
library(wordcloud)
```

```
library(RColorBrewer)

# Sample Customer Feedback
feedback <- c(
  "Excellent service and friendly staff",
  "Very satisfied with product quality",
  "Fast delivery and excellent support",
  "Good experience and helpful employees",
  "Satisfied with service and quick response"
)

# Create Word Cloud
wordcloud(
  words = feedback,
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2"),
  scale = c(4,0.8)
)
```



SET 3:

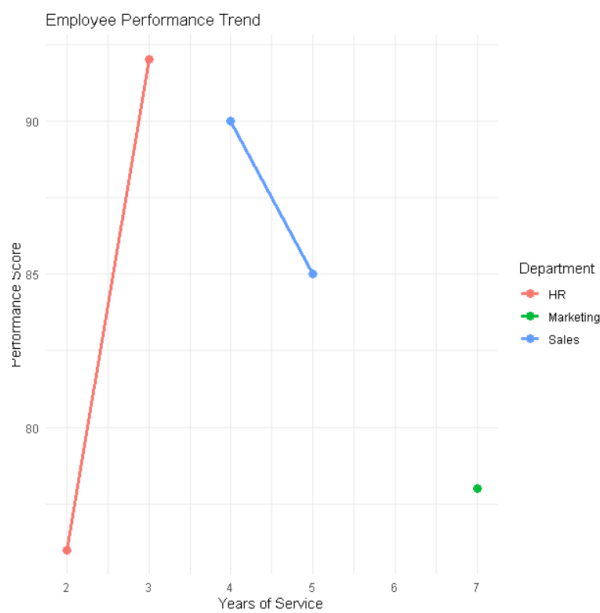
Question 1: Line Chart - Employee Performance Trend

R program:

```
library(ggplot2)

employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsService = c(5,3,7,4,2),
  PerformanceScore = c(85,92,78,90,76)
)

ggplot(employee,
  aes(x = YearsService,
      y = PerformanceScore,
      color = Department,
      group = Department)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score",
    color = "Department"
  ) +
  theme_minimal()
```



Question 2: Bar Chart - Employees by Department

R program:

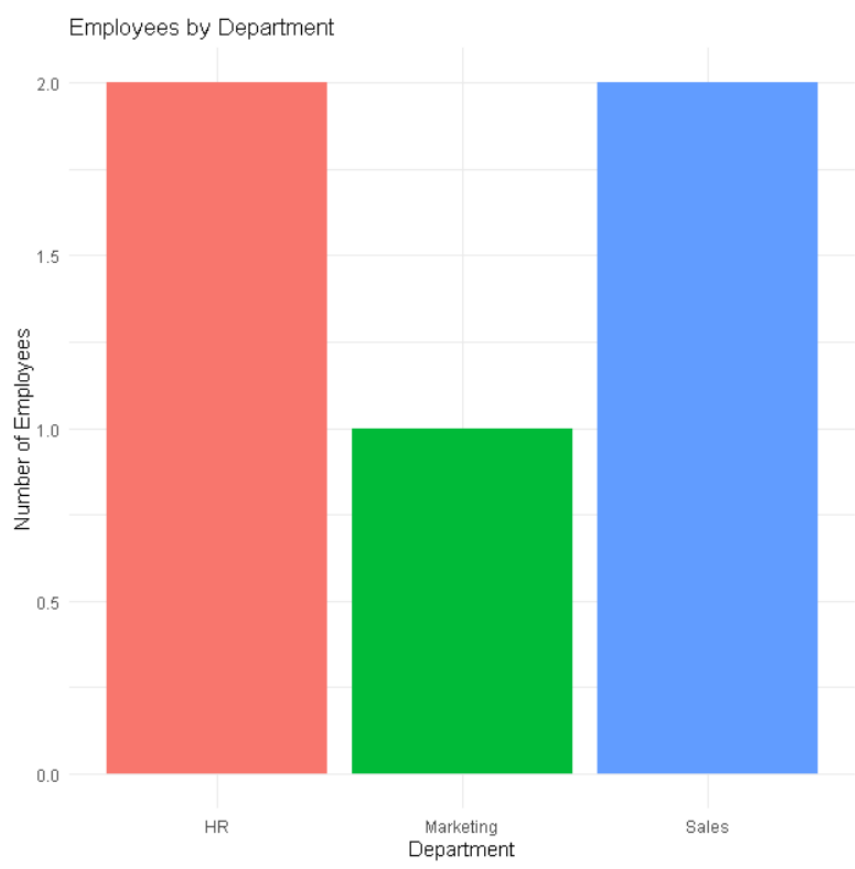
```
library(ggplot2)

# Dataset

employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsService = c(5,3,7,4,2),
  PerformanceScore = c(85,92,78,90,76)
)

ggplot(employee,
  aes(x = Department,
    fill = Department)) +
  geom_bar() +
  labs(
    title = "Employees by Department",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal() +
```

```
theme(legend.position = "none")
```



Question 3: Scatter Plot - Years of Service vs Performance Score

R program:

```
library(ggplot2)

# Dataset

employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsService = c(5,3,7,4,2),
  PerformanceScore = c(85,92,78,90,76)
)

ggplot(employee,
  aes(x = YearsService,
      y = PerformanceScore,
```

```

    color = Department)) +
  geom_point(size = 4) +
  geom_smooth(method = "lm",
    se = FALSE,
    color = "black") +
  labs(
    title = "Years of Service vs Performance Score",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()

```



Question 4: Interactive Dashboard

R Program:

```

library(shiny)
library(plotly)
employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),

```

```

Department = c("Sales","HR","Marketing","Sales","HR"),
YearsService = c(5,3,7,4,2),
PerformanceScore = c(85,92,78,90,76)
)
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  fluidRow(
    column(6,
      plotlyOutput("linePlot")),
    column(6,
      plotlyOutput("barPlot"))
  )
)
server <- function(input, output){
  output$linePlot <- renderPlotly({
    plot_ly(
      employee,
      x = ~YearsService,
      y = ~PerformanceScore,
      color = ~Department,
      type = "scatter",
      mode = "lines+markers"
    )
  })
  output$barPlot <- renderPlotly({
    dept <- as.data.frame(table(employee$Department))
    names(dept) <- c("Department","Count")
    plot_ly(
      dept,
      x = ~Department,

```



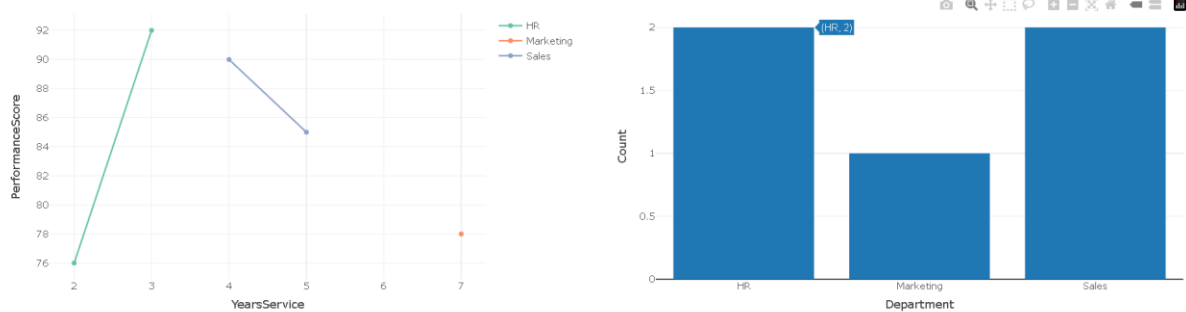
```

    y = ~Count,
    type = "bar"
  )
})
}

shinyApp(ui, server)

```

Employee Performance Dashboard



SET -4

Question 1: Bar Chart - Quantity of Each Product

```

library(ggplot2)

inventory <- data.frame(
  ProductID = c(1,2,3,4,5),
  ProductName = c("Product A","Product B","Product C","Product D","Product E"),
  Quantity = c(250,175,300,200,220)
)

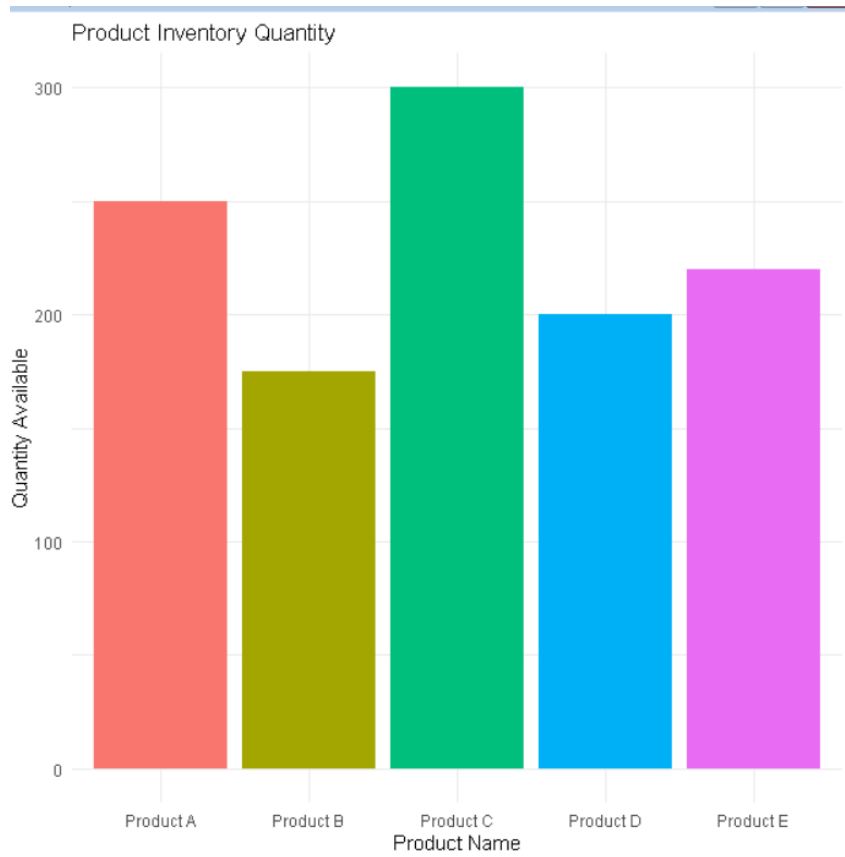
ggplot(inventory,
  aes(x = ProductName,
    y = Quantity,
    fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Product Inventory Quantity",

```

```

x = "Product Name",
y = "Quantity Available"
) +
theme_minimal() +
theme(legend.position = "none")

```



Question 2: Stacked Bar Chart - Product Categories

R program:

```

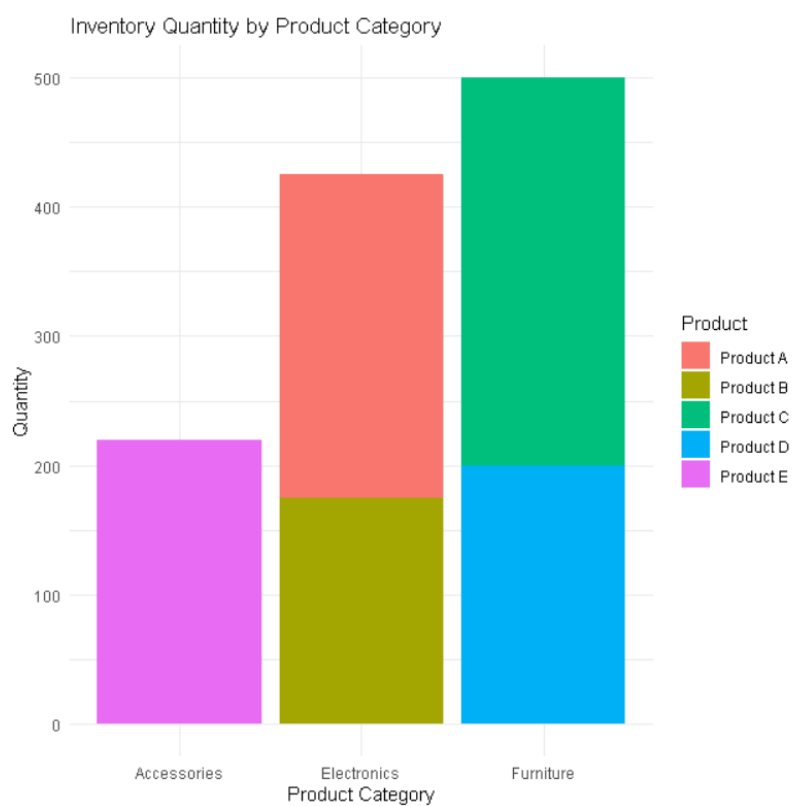
library(ggplot2)
inventory <- data.frame(
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Accessories"),
  Quantity = c(250, 175, 300, 200, 220)
)
ggplot(inventory,
  aes(x = Category,
    y = Quantity,

```

```

    fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Inventory Quantity by Product Category",
    x = "Product Category",
    y = "Quantity",
    fill = "Product"
  ) +
  theme_minimal()

```



Question 3: Scatter Plot - Product Price vs Quantity

R Program:

```

library(ggplot2)

inventory <- data.frame(
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Price = c(500, 350, 700, 450, 600),
  Quantity = c(250, 175, 300, 200, 220)
)

```

```

ggplot(inventory,
       aes(x = Price,
           y = Quantity)) +
  geom_point(color = "blue",
            size = 4) +
  geom_smooth(method = "lm",
            color = "red",
            se = FALSE) +
  labs(
    title = "Product Price vs Quantity Available",
    x = "Product Price",
    y = "Quantity Available"
  ) +
  theme_minimal()

```



Question 4: Interactive Dashboard (Shiny + Plotly)

R Program:

```
library(shiny)
library(plotly)

inventory <- data.frame(
  ProductName = c("Product A","Product B","Product C","Product D","Product E"),
  Quantity = c(250,175,300,200,220),
  Category = c("Electronics","Electronics","Furniture","Furniture","Accessories")
)

ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  fluidRow(
    column(6,
      plotlyOutput("barPlot")),
    column(6,
      plotlyOutput("stackPlot"))
  )
)

server <- function(input, output){
  output$barPlot <- renderPlotly({
    plot_ly(
      inventory,
      x = ~ProductName,
      y = ~Quantity,
      type = "bar",
      color = ~ProductName
    )
  })

  output$stackPlot <- renderPlotly({
    plot_ly(
      inventory,
```

```

x = ~Category,
y = ~Quantity,
color = ~ProductName,
type = "bar"
) %>%
  layout(barmode = "stack")
})
}
shinyApp(ui, server)

```



SET-5

Question 1: Line Chart - Daily Page Views

R Program:

```

library(ggplot2)

website <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
    "2023-01-04", "2023-01-05")),
  PageViews = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

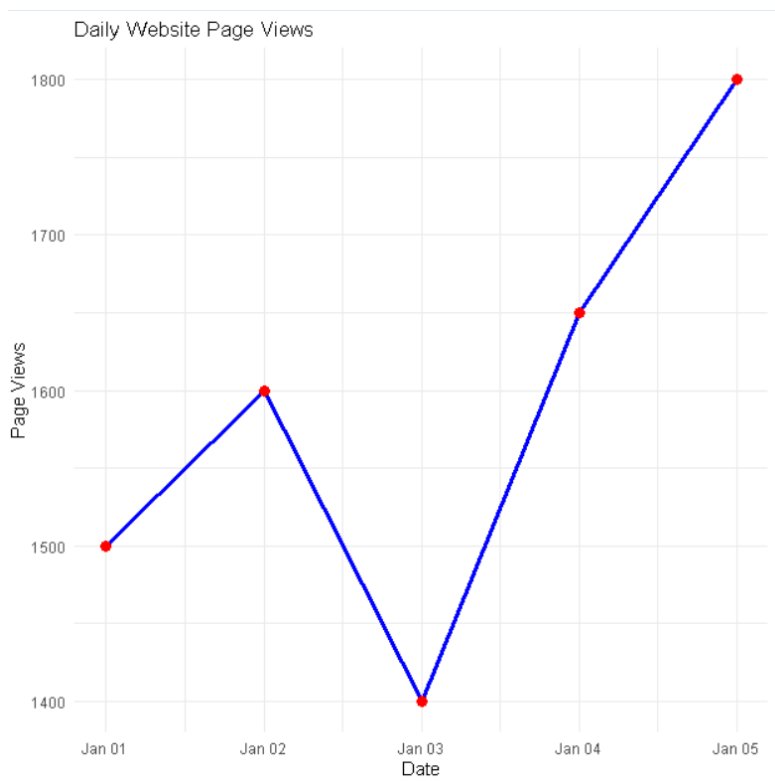
ggplot(website,
  aes(x = Date,

```

```

    y = PageViews,
    group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Website Page Views",
    x = "Date",
    y = "Page Views"
  ) +
  theme_minimal()

```



Question 2: Bar Chart - Top Click-Through Rates

RProgram

```

library(ggplot2)

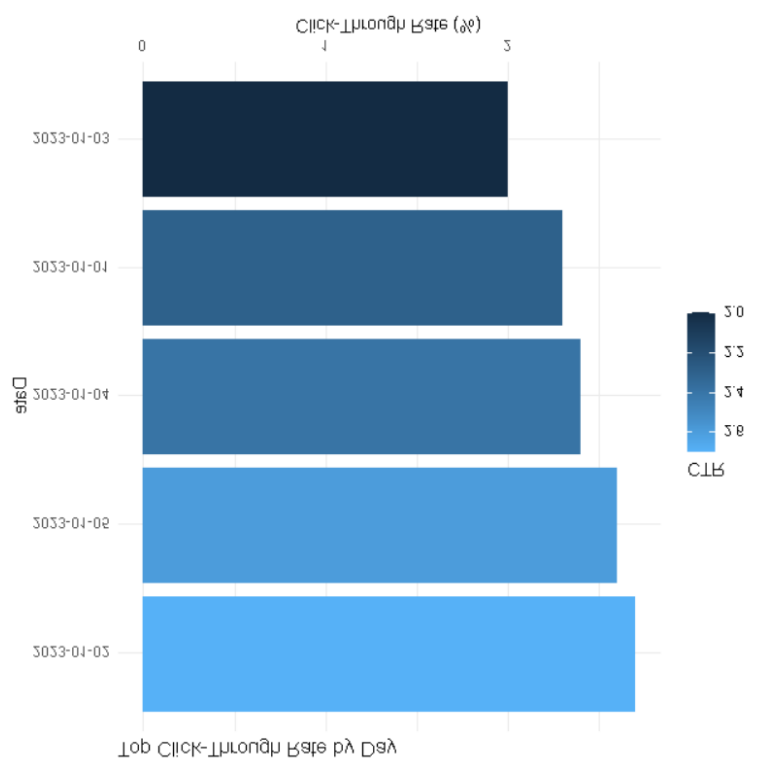
website <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
    "2023-01-04", "2023-01-05")),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

```

```

)
ggplot(website,
  aes(x = reorder(as.character(Date), CTR),
    y = CTR,
    fill = CTR)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(
    title = "Top Click-Through Rate by Day",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal()

```



Question 3: Stacked Area Chart - User Interactions

R PROGRAM

```
library(ggplot2)
```



```

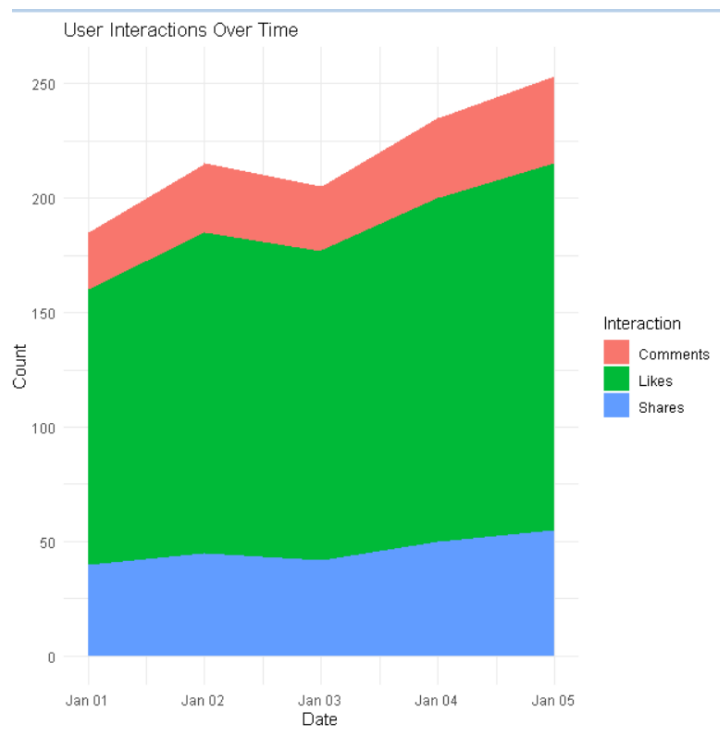
interaction <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
    "2023-01-04", "2023-01-05")),
  Likes = c(120, 140, 135, 150, 160),
  Shares = c(40, 45, 42, 50, 55),
  Comments = c(25, 30, 28, 35, 38)
)

library(tidyr)

interaction_long <- pivot_longer(
  interaction,
  cols = Likes:Comments,
  names_to = "Interaction",
  values_to = "Count"
)

ggplot(interaction_long,
  aes(x = Date,
    y = Count,
    fill = Interaction)) +
  geom_area() +
  labs(
    title = "User Interactions Over Time",
    x = "Date",
    y = "Count"
  ) +
  theme_minimal()

```



Question 4: Interactive Dashboard (Shiny + Plotly)

R Program:

```
library(shiny)
library(plotly)

website <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
    "2023-01-04", "2023-01-05")),
  PageViews = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)

ui <- fluidPage(
  titlePanel("Website Analytics Dashboard"),
  fluidRow(
    column(
      6,
      plotlyOutput("linePlot")
    ),
    column(
```

```

        6,
        plotlyOutput("barPlot")
    )
)
)
server <- function(input, output){
  output$linePlot <- renderPlotly({
    plot_ly(
      website,
      x = ~Date,
      y = ~PageViews,
      type = "scatter",
      mode = "lines+markers"
    )
  })
  output$barPlot <- renderPlotly({
    plot_ly(
      website,
      x = ~Date,
      y = ~CTR,
      type = "bar",
      color = ~CTR
    )
  })
}
shinyApp(ui, server)

```

Website Analytics Dashboard

